

STEP 1 — Install and Import Required Libraries

```
# Install Hugging Face Transformers and supporting libraries
!pip install transformers
!pip install torch
!pip install sentencepiece
# Import summarization pipeline from transformers
from transformers import pipeline
import textwrap
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.6.0)
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Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggi
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0-
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,
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Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch) (3.0.3)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
```

STEP 2 — Load Summarization Model

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
```

```
# Load tokenizer and model explicitly for sequence-to-sequence tasks (like summarization)
tokenizer = AutoTokenizer.from_pretrained("facebook/bart-large-cnn")
model = AutoModelForSeq2SeqLM.from_pretrained("facebook/bart-large-cnn")
```

```
print("Summarization model and tokenizer loaded successfully!")
```

```
Loading weights: 100%
```

```
511/511 [00:02<00:00, 260.22it/s, Materializing param=model.encoder.layers.11.self_attn_layer_norm.weight]
```

```
Summarization model and tokenizer loaded successfully!
```

STEP 3 — Prepare Input Texts

```
texts = [
```

```
    """Artificial Intelligence is transforming industries worldwide.
    Companies use AI to automate processes, analyze large datasets,
    and improve decision making. Machine learning models help detect
    patterns in data and provide predictions. AI technologies are
    also widely used in healthcare, finance, transportation, and
    education. However, ethical concerns such as bias and privacy
```

```

remain important challenges in AI development.""",

"""The Indian Space Research Organisation launched several
successful satellite missions in recent years. These satellites
help improve communication, weather forecasting, and disaster
management. India has also achieved significant milestones in
planetary exploration missions such as Mars Orbiter Mission
and Chandrayaan.""",

"""Once upon a time a small village faced severe drought.
Farmers struggled to grow crops due to lack of rainfall.
One day heavy rains arrived bringing hope and prosperity
to the farmers. The village celebrated the return of water
and began rebuilding their farms."""
]

```

STEP 4 — Generate Summaries

```

for text in texts:
    # Tokenize input
    inputs = tokenizer([text], max_length=1024, return_tensors='pt', truncation=True)

    # Generate summary
    summary_ids = model.generate(inputs['input_ids'], num_beams=4, max_length=60, min_length=20, early_stopping=True)

    # Decode summary
    summary_text = [tokenizer.decode(g, skip_special_tokens=True, clean_up_tokenization_spaces=False) for g in summary_ids]

    # Wrap the summary text for paragraph display
    wrapped_summary = textwrap.fill(summary_text[0], width=80) # Adjust width as needed

    print("\nOriginal Text:\n", text)
    print("\nSummary:\n", wrapped_summary)

```

Original Text:
 Artificial Intelligence is transforming industries worldwide. Companies use AI to automate processes, analyze large datasets, and improve decision making. Machine learning models help detect patterns in data and provide predictions. AI technologies are also widely used in healthcare, finance, transportation, and education. However, ethical concerns such as bias and privacy remain important challenges in AI development.

Summary:
 Artificial Intelligence is transforming industries worldwide. Machine learning models help detect patterns in data and provide predictions. ethical concerns such as bias and privacy remain important challenges.

Original Text:
 The Indian Space Research Organisation launched several successful satellite missions in recent years. These satellites help improve communication, weather forecasting, and disaster management. India has also achieved significant milestones in planetary exploration missions such as Mars Orbiter Mission and Chandrayaan.

Summary:
 The Indian Space Research Organisation launched several successful satellite missions in recent years. These satellites help improve communication, weather forecasting, and disaster management. India has also achieved significant milestones in planetary exploration missions such as Mars Orbiter Mission and Chandrayaan.

Original Text:
 Once upon a time a small village faced severe drought. Farmers struggled to grow crops due to lack of rainfall. One day heavy rains arrived bringing hope and prosperity to the farmers. The village celebrated the return of water and began rebuilding their farms.

Summary:
 A small village faced severe drought due to lack of rainfall. One day heavy rains arrived bringing hope and prosperity to the farmers.

STEP 5 — Control Summary Length

```
text = texts[0]
```

```
# Tokenize input
inputs = tokenizer([text], max_length=1024, return_tensors='pt', truncation=True)
input_ids = inputs['input_ids']

# Generate short summary
short_summary_ids = model.generate(input_ids, num_beams=4, max_length=30, min_length=10, early_stopping=True)
short_summary = tokenizer.decode(short_summary_ids[0], skip_special_tokens=True, clean_up_tokenization_spaces=False)

# Generate medium summary
medium_summary_ids = model.generate(input_ids, num_beams=4, max_length=60, min_length=20, early_stopping=True)
medium_summary = tokenizer.decode(medium_summary_ids[0], skip_special_tokens=True, clean_up_tokenization_spaces=False)

# Generate long summary
long_summary_ids = model.generate(input_ids, num_beams=4, max_length=100, min_length=40, early_stopping=True)
long_summary = tokenizer.decode(long_summary_ids[0], skip_special_tokens=True, clean_up_tokenization_spaces=False)

print("Short Summary:\n", short_summary)
print("\nMedium Summary:\n", medium_summary)
print("\nLong Summary:\n", long_summary)
```

```
Short Summary:
Artificial Intelligence is transforming industries worldwide. Machine learning models help detect patterns in data and provide p

Medium Summary:
Artificial Intelligence is transforming industries worldwide. Machine learning models help detect patterns in data and provide p

Long Summary:
Artificial Intelligence is transforming industries worldwide. Machine learning models help detect patterns in data and provide p
```

STEP 6 — Compare Different Text Types (Analysis Example)

The summarization model performs differently depending on the type of input text. News articles are usually summarized well because the model is trained on similar structured content. Technical texts may lose some domain-specific terms because the model tries to simplify complex explanations. Narrative texts are summarized by capturing the main storyline but may omit descriptive details. Conversational text can sometimes produce less coherent summaries due to informal language patterns. Information coverage is generally good when the text contains clear topic sentences. However, some minor details are often removed during the summarization process. The model maintains coherence in most summaries, producing grammatically correct sentences. In certain cases, key technical information may be partially lost. Overall, the model performs best on structured informational text such as news articles.

STEP 7 — Test with Very Long Text

```
long_text = """
Artificial Intelligence has rapidly evolved over the past decade.
Organizations across industries are integrating AI systems to
improve productivity and decision making. Machine learning
algorithms can analyze massive datasets and identify patterns
that humans may not detect easily. These capabilities are used
in applications such as medical diagnosis, financial fraud
detection, recommendation systems, and autonomous vehicles.
```

```
Despite these benefits, AI development raises ethical concerns.
Issues such as algorithmic bias, data privacy, and transparency
are widely discussed among researchers and policymakers.
Ensuring responsible AI development is essential to avoid
harmful consequences.
```

```
Governments and organizations are working to establish
regulations and ethical guidelines for AI deployment.
Educational institutions are also introducing AI-related
courses to prepare students for the future workforce.
```

```
As AI technology continues to advance, it is expected to
transform many aspects of daily life, creating both
opportunities and challenges for society.
"""
```

```
# Tokenize input
inputs = tokenizer([long_text], max_length=1024, return_tensors='pt', truncation=True)
input_ids = inputs['input_ids']
```

```
# Generate summary
summary_ids = model.generate(input_ids, num_beams=4, max_length=120, min_length=50, early_stopping=True)
summary = tokenizer.decode(summary_ids[0], skip_special_tokens=True, clean_up_tokenization_spaces=False)

print(summary)
```

Artificial Intelligence has rapidly evolved over the past decade. Machine learning can analyze massive datasets and identify pat

STEP 8 — Lab Report Section

The lab implements text summarization using pretrained Transformer models from the Hugging Face library on a dataset containing news, technical, and narrative texts, where libraries such as Transformers, Torch, and SentencePiece are installed, the BART summarization pipeline generates summaries capturing main ideas, parameters like max_length and min_length are compared to produce summaries of different lengths, analysis shows better performance on structured texts like news while sometimes losing technical details, and the conclusion states that Transformer models produce efficient and coherent summaries though they may struggle with domain-specific information and very long texts.